Venkata Lakshmi Parimala Pasupuleti

AI/ML Engineer · Python / LangGraph / AWS Bedrock & SageMaker · LLMs & Generative AI · MLOps & Fine-tuning

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SUMMARY

AI/ML Engineer with 9 years of production software ownership. Built AgentProbe -- an open-source LLMOps and MLOps platform: CI-integrated eval gate, LLM-as-judge scoring, hallucination detection, and model monitoring. Built LeaseI -- an 8-agent LangGraph system with RAG, fine-tuning workflows, prompt engineering, and HITL guardrails on regulated decisions, evaluated across 1,240 test cases at 93.6% precision. Model development: fine-tuned a custom classification architecture using HuggingFace Transformers and PyTorch, deployed on Quart. AWS across full SDLC (S3, EC2, RDS, Glue, Bedrock, SageMaker, Lambda); GCP and Azure cloud experience. Seven years shipping enterprise Python systems at scale with HIPAA compliance. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI/ML & LLM Engineering: Python, LangGraph (LangGraph), LangChain, Agentic AI, Multi-Agent Orchestration, Large Language Models (LLMs), Generative AI, Fine-tuning (HuggingFace Transformers, PyTorch), RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Prompt Engineering, Tool Calling, HITL, Guardrails, Output Validation, Hallucination Detection, Structured Outputs, NLP, BERT, spaCy, TensorFlow, scikit-learn

MLOps & Cloud: AWS (Bedrock, SageMaker, S3, EC2, RDS, Lambda, Glue), Google Cloud Platform (GCP), Azure, MLOps, LLMOps, Model Deployment, Model Monitoring, CI/CD (GitHub Actions), Docker, Kubernetes (K8s), FastAPI, REST APIs

Data & Infrastructure: PostgreSQL, MySQL, Oracle, RabbitMQ, Redis, Microservices, Data Pipelines, ETL, HIPAA Compliance, PHI Data Handling, Data Privacy, Libsodium, Git, Linux, Agile/Scrum

Languages: Python, JavaScript, TypeScript, Java, SQL, PL/SQL, React.js, Angular, Flask, Bash, HTML/CSS

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a production LangGraph agentic AI system with RAG retrieval, prompt-engineered guardrails, and LLM output validation for an external client; owned design through delivery including model selection, fine-tuning strategy, and integration with enterprise data sources.

Entrata (PropTech SaaS · 100K+ MAU · PHI compliance · enterprise systems)

Sep 2018 - Aug 2024

Senior Software Engineer

- Shipped 20+ full-stack production modules (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; built scalable data pipelines and APIs supporting enterprise workflows at 100K+ MAU scale.
- Owned compliance-critical data systems (180,000 HUD records, housing eligibility for 100K+ residents) -- zero violations across 7 years; baseline validation and rollback on every release cycle.
- Led AWS S3 document migration with full data privacy and HIPAA compliance -- Libsodium-encrypted pipelines, zero security incidents; built and maintained Python/Bash automation pipelines eliminating 80% of manual workload.
- Optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers; conducted 300+ peer code reviews; reduced debugging time by 40% through internal tooling.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Fine-tuned a custom classification model architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented model design, training methodology, and evaluation results to faculty.
- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest for automated data integrity checks across 3 research pipelines.

PROJECTS

AgentProbe: MLOps & LLM Evaluation Platform

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing

- Open-source MLOps and LLM evaluation platform: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, model monitoring, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama). 5-metric eval engine; 15-endpoint FastAPI backend; evaluated LeaseI across 1,240+ production-quality test cases.

LeaseI: 8-Agent LangGraph Production AI System

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Production agentic AI system: 8 LangGraph agents with RAG, prompt engineering, structured data extraction, HITL guardrails on regulated decisions, and conditional routing. 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: HIPAA-Compliant Healthcare AI System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- HIPAA-compliant AI system over structured PHI data -- RBAC, data privacy controls, LLM + BERT + embeddings integration; reduced clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017